

BOUNDED EXCEPTIONAL ZEROS FOR POLYNOMIAL INITIAL CONDITIONS IN THE FORGÁCS–TRAN FAMILY

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ABSTRACT. Let $Q \in \mathbb{R}[t]$ be nonconstant with $Q(0) > 0$ and only positive real zeros, fix $r \geq 1$ with $\max\{\deg Q, r\} > 1$ and a numerator $N \in \mathbb{R}[t, z] \setminus \{0\}$ with $\deg_t N < \deg_t(Q + zt^r)$, and define $P_m(z)$ by $N(t, z)/(Q(t) + zt^r) = \sum_{m \geq 0} P_m(z)t^m$. The denominator determines a linear recurrence and N its polynomial initial conditions; for $N = 1$, Forgács and Tran proved every sufficiently large P_m real-rooted, with all zeros in an explicit interval $I_{Q,r} \subset (0, \infty)$. They ask whether the number of zeros of P_m outside $(0, \infty)$ admits a bound depending only on r and $\deg Q$. We prove that for each fixed N there is a constant $C = C(Q, r, N)$ such that every nonzero P_m has at most C zeros outside $(0, \infty)$, counted with multiplicity, and every sufficiently large P_m at most C outside $I_{Q,r}$. The latter have at least $\deg P_m - C$ distinct zeros in $I_{Q,r}$, and $\deg P_m = m/r + O_{Q,r,N}(1)$, so arbitrary fixed polynomial initial conditions cost the positive-real bulk only a bounded defect. Normalized by degree, the zero-counting measures converge weakly to the pushforward of uniform angular measure under the Forgács–Tran parameterization, the same limit for every fixed numerator. Neither the defect nor the dependence of C on N can be removed: a numerator $R(z)$ forces arbitrarily many negative, hence exceptional, zeros into every nonzero P_m .

1. INTRODUCTION

The zeros of polynomial sequences defined by rational generating functions are often controlled by the relative moduli of the denominator roots. That their limiting loci are governed by competition between dominant denominator roots and by the vanishing of the associated residue amplitudes goes back to Beraha, Kahane, and Weiss [2] and, in the general form of competing analytic branches $\sum_k \alpha_k(z)\beta_k(z)^m$, to Sokal [10, Thm. 1.5]; for coefficient sequences of rational generating functions it was developed by Tran [11, 12]. Forgács and Tran introduced the following positive-rooted family. Let $Q \in \mathbb{R}[t]$ satisfy

$$(1.1) \quad Q(0) > 0, \quad Q(t) = Q(0) \prod_{j=1}^n \left(1 - \frac{t}{x_j}\right), \quad 0 < x_1 \leq \cdots \leq x_n,$$

where multiplicities are allowed, and let $r \geq 1$. The denominator-only sequence

$$(1.2) \quad \frac{1}{Q(t) + zt^r} = \sum_{m \geq 0} H_m(z)t^m$$

was shown by Forgács and Tran to be eventually hyperbolic—every sufficiently large H_m real-rooted—with all zeros lying in an explicitly determined interval $I_{Q,r} = (a, b) \subset (0, \infty)$ [5]. Their proof constructs a strictly monotone parameterization $z = z(\theta)$, $0 < \theta < \pi/r$, for which the two minimum-modulus denominator zeros form the conjugate pair

$$t_{\pm}(\theta) = \tau(\theta)e^{\pm i\theta}.$$

This pair is called *principal*, the remaining zeros of the denominator *nonprincipal*. The coefficient $H_m(z(\theta))$ is then governed by the interference of these two modes. Estimates near both

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endpoints show that the resulting phase produces at least $\lfloor m/r \rfloor$ positive zeros, the maximum permitted by the degree.

In Problem 14 of their earlier paper [4], Forgács and Tran asked what remains true when the numerator is allowed to vary. There one fixes a real bivariate polynomial $N(t, z)$ with

$$\deg_t N < \deg_t(Q(t) + zt^r)$$

and defines

$$(1.3) \quad \frac{N(t, z)}{Q(t) + zt^r} = \sum_{m \geq 0} P_m(z) t^m.$$

They ask whether the number of zeros of P_m outside $(0, \infty)$ is bounded independently of m . As printed, the constant depends at most on r and $\deg Q$, hence must be uniform over the numerator. The surrounding discussion, however, treats the numerator as the fixed initial conditions for the denominator recurrence: there “the denominator of the generating function gives the recurrence relation for $P_m(z)$ whereas the numerator gives rise to the set of initial polynomials”, an equivalence made precise in Proposition 2.1. On the same page the problem is also rephrased as showing that the P_m have zero loci similar to those of the sequence generated by $1/(Q(t) + zt^r)$, after [7]; that gloss is the asymptotic reading, settled by Proposition 6.1. The theorem below treats the fixed-numerator regime, with N held fixed as $m \rightarrow \infty$ rather than varying with the index.

The dependence on N cannot be omitted. If $R(z) \in \mathbb{R}[z]$ and $N(t, z) = R(z)$, then

$$P_m(z) = R(z)H_m(z),$$

so the fixed zeros of R occur in every coefficient polynomial. Any bound on the exceptional zeros must therefore depend on N , giving $C = C(Q, r, N)$ and showing that no uniform-in- N bound is possible; see Proposition 6.3.

The question is one of stability under a change of initial conditions. The denominator $Q(t) + zt^r$ fixes the recurrence and its pole geometry, while N fixes the initial conditions; eventual hyperbolicity for the single numerator $N = 1$ need not persist when that data changes. Theorem 1.1 shows that it persists with only a bounded loss: arbitrary fixed polynomial initial conditions can create exceptional zeros, but only $O(1)$ of them. The asymptotic positive-real zero bulk and its containing interval are therefore determined by the denominator alone. The normalized zero-counting measures converge weakly to a single limit, the same for every fixed numerator (Proposition 6.1). Beraha–Kahane–Weiss theory [2, 10] identifies the loci where the zeros accumulate but gives no bound on the number lying away from them at finite degree; the present theorem supplies that bound, uniformly in m .

Tran noted that his zero-location theorem for polynomials with a three-term recurrence persists for a monomial numerator in t and z , and reported unpublished work with Boyer that a general numerator sends the zeros toward the curve of that theorem up to a possible finite exceptional set [12, p. 879]. That indication is asymptotic, a statement about the limiting zero set, whereas Theorem 1.1 bounds the exceptional zeros of each P_m uniformly in m , for the positive-rooted pencil and with the explicit interval $I_{Q,r}$.

For polynomials satisfying a three-term recurrence, Mai bounded the number of zeros that leave the recurrence’s limiting curve under initial conditions more general than the canonical one: for the family $(aB(z)t + b)/(1 + B(z)t + A(z)t^2)$, whose letters are his and unrelated to those used here, he obtained at most $\max\{\deg A, 2 \deg B\}$ such zeros, uniformly in the index [9, Thm. 1]. The same phenomenon, finitely many exceptional zeros created by a change of initial conditions, occurs here in a complementary setting. There the recurrence is of second order and the initial conditions available there form the two-parameter family $P_0 = b$, $P_1 = (a - b)B$, and the limiting locus may be a complex curve; here the denominator is the positive-rooted pencil $Q(t) + zt^r$ of arbitrary order, the initial conditions are arbitrary (Proposition 2.1), and the locus is the real interval $I_{Q,r}$.

Bounds of this kind are known for other fixed pencils. For the doubly indexed family generated by $\tilde{N}(s, t, z)/(1 + s + t + zst)$, where the numerator likewise supplies the initial polynomials,

Luong and Tran bounded the number of nonreal zeros by $\deg_s \tilde{N} + \deg_t \tilde{N} + 2 \deg_z \tilde{N}$ [8]. Al Hamdani and Tran treated $1/((1-t)^\alpha(1-2zt+t^2))$ with $0 < \alpha < 1$ and bounded the number of zeros outside $(-1, 1)$ by a constant independent of the index [1], by a conjugate-pair extraction and phase count of the same shape as the argument below. Neither denominator is of the form $Q(t) + zt^r$ with Q positive-rooted, and neither bound contains or is contained in Theorem 1.1: the pencil fixed here is the one of Problem 14, and the zeros are counted off the positive ray rather than off $\mathbb{R}$ or off a fixed interval.

The main theorem is the following.

Theorem 1.1 (Fixed-numerator Forgács–Tran theorem). *Let $Q \in \mathbb{R}[t]$ be nonconstant and satisfy (1.1). Let $r \geq 1$ and assume*

$$\max\{\deg Q, r\} > 1.$$

Let $N \in \mathbb{R}[t, z] \setminus \{0\}$ satisfy

$$\deg_t N < \deg_t(Q(t) + zt^r),$$

where $\deg_t$ is the degree in t over $\mathbb{R}[z]$, so that $\deg_t(Q(t) + zt^r) = \max\{\deg Q, r\}$. Define $P_m \in \mathbb{R}[z]$ by (1.3). Then there exists $C = C(Q, r, N)$ such that every nonzero P_m has at most C zeros outside $(0, \infty)$, counted with multiplicity.

More precisely, if $I_{Q,r} = (a, b)$ is the Forgács–Tran interval defined in (3.1), then every sufficiently large P_m is nonzero, has at most C zeros outside $I_{Q,r}$, counted with multiplicity, and has at least $\deg P_m - C$ distinct zeros in $I_{Q,r}$.

Brändén and Leite have since proved all-degree real-rootedness and interlacing for the denominator-only family through total nonnegativity [3, Thm. 4.5], settling Problem 13 of Forgács and Tran [4] for $r \geq 1$. Their parameter is the negative of z , and the zeros they obtain are negative, so for every Q satisfying (1.1) the conclusion is real-rootedness together with containment in $(0, \infty)$ rather than real-rootedness alone. That family is the case $N = 1$; the present theorem concerns arbitrary fixed initial conditions, where the numerator may force nonreal or nonpositive zeros. The conclusion here is that this failure is confined to a bounded defect.

The excluded case $\deg Q = r = 1$ is elementary and satisfies the first conclusion of Theorem 1.1, the second being vacuous there; see Proposition 5.2. The hypothesis $r \geq 1$ is essential to the polynomial formulation: for $r = 0$, the coefficients are generally rational rather than polynomial functions of z . We also exclude constant Q , for which P_m may have a zero of large multiplicity at the origin. Apart from Proposition 5.2, the hypotheses (1.1) and $\max\{\deg Q, r\} > 1$ are in force throughout; a denominator $Q(t) + zt^r$ meeting them is called *admissible*. The latter gives $rQ(t) - tQ'(t)$ a positive zero, hence t_a , and at $r = 1$ forces $\deg Q \geq 2$ and with it the negative zero t_b , so that $I_{Q,r}$ exists; at $\deg Q = r = 1$ that polynomial reduces to the nonzero constant $Q(0)$.

The proof has four components. Section 2 reduces, by an exact division in the Laurent polynomial ring, an arbitrary bivariate numerator to a fixed univariate numerator $B(t)$ with $B(0) \neq 0$ at the cost of a finite index shift $m \mapsto M = m + O(1)$, and establishes the eventual degree formula

$$\deg[t^M] \frac{B(t)}{Q(t) + zt^r} = \left\lfloor \frac{M}{r} \right\rfloor.$$

Section 3 records the denominator geometry imported from Forgács and Tran and the principal-pair decomposition; the univariate numerator enters only through the amplitude $B(t_+(\theta))$, which has finitely many zeros on the branch and phase of bounded variation (Lemma 3.6).

Section 4 extends the three-region estimates of [5, Prop. 3] to this weighted amplitude: away from the two endpoints and from small neighborhoods of the amplitude zeros, the principal conjugate pair dominates, and truncating angular windows of width $O(1/M)$ costs only $O(1)$ phase oscillations. Section 5 counts the sign changes of the resulting phase, giving $\lfloor M/r \rfloor - O(1)$ distinct zeros in $I_{Q,r}$; the eventual degree formula completes the proof.

2. LAURENT REDUCTION AND EVENTUAL DEGREE

Throughout the paper, put

$$D(t, z) = Q(t) + zt^r, \quad d = \deg_t D = \max\{\deg Q, r\}.$$

The numerator carries the same information as the initial conditions of the denominator recurrence.

Proposition 2.1. *Write $D(t, z) = \sum_{j=0}^d d_j(z)t^j$, so that $d_0 = Q(0) \neq 0$. For each $N = \sum_{m=0}^{d-1} N_m(z)t^m$ with $N_m \in \mathbb{R}[z]$, the coefficients P_m of N/D satisfy*

$$\sum_{j=0}^m d_j(z)P_{m-j}(z) = N_m(z) \quad (0 \leq m < d).$$

This is a lower-triangular system with constant diagonal $d_0 = Q(0)$, so $N \mapsto (P_0, \dots, P_{d-1})$ is a bijection from proper numerators, those with $\deg_t N < d$, onto $\mathbb{R}[z]^d$. A fixed proper N is thus the same datum as a fixed vector $(P_0, \dots, P_{d-1})$ of polynomial initial conditions.

Proof. Equating coefficients of t^m in $N = D \sum_{k \geq 0} P_k t^k$ gives the displayed equations for $0 \leq m < d$ and $\sum_{j=0}^d d_j(z)P_{m-j}(z) = 0$ for $m \geq d$. The first d equations express $(N_0, \dots, N_{d-1})$ from $(P_0, \dots, P_{d-1})$ by a lower-triangular matrix with constant diagonal $d_0 = Q(0) \neq 0$, invertible over $\mathbb{R}[z]$; the remaining equations then determine P_m for $m \geq d$. $\square$

The following reduction uses the linearity of D in z together with the fact that its z -coefficient t^r is a unit in the Laurent polynomial ring $\mathbb{R}[t, t^{-1}]$.

Lemma 2.2 (Laurent–Gauss reduction). *Let $N \in \mathbb{R}[t, z] \setminus \{0\}$ satisfy $\deg_t N < d$, and set $E = \deg_z N$. Define*

$$(2.1) \quad A(t) = t^{rE} N\left(t, -\frac{Q(t)}{t^r}\right).$$

Then $A \in \mathbb{R}[t] \setminus \{0\}$. There is an $S \in \mathbb{R}[t, t^{-1}, z]$ such that

$$(2.2) \quad N(t, z) = D(t, z)S(t, z) + t^{-rE}A(t).$$

If $A(t) = t^\mu B(t)$ with $B(0) \neq 0$, then for all sufficiently large m ,

$$(2.3) \quad P_m(z) = [t^{m+rE-\mu}] \frac{B(t)}{D(t, z)}.$$

Proof. Write

$$N(t, z) = \sum_{\alpha, \beta} c_{\alpha, \beta} t^\alpha z^\beta.$$

After the substitution in (2.1), the corresponding term is

$$c_{\alpha, \beta} (-1)^\beta t^{\alpha+r(E-\beta)} Q(t)^\beta,$$

which is a polynomial in t . Thus $A \in \mathbb{R}[t]$.

Suppose that $A \equiv 0$. Then, as a polynomial in z over the field $\mathbb{R}(t)$,

$$N\left(t, -\frac{Q(t)}{t^r}\right) = 0.$$

Since $D(t, z)$ is linear in z , it follows that D divides N in $\mathbb{R}(t)[z]$. The polynomials $Q(t)$ and t^r are coprime because $Q(0) \neq 0$, so D is primitive in $\mathbb{R}[t][z]$. Gauss's lemma gives $D \mid N$ in $\mathbb{R}[t, z]$. Since $\mathbb{R}[z][t]$ is an integral domain,

$$\deg_t N \geq \deg_t D = d,$$

a contradiction. Hence $A \neq 0$.

In the Laurent ring $\mathbb{R}[t, t^{-1}][z]$, the polynomial

$$N(t, z) - t^{-rE}A(t)$$

vanishes at $z = -Q(t)/t^r$. Since $z + Q(t)/t^r$ is monic in z , the factor theorem gives divisibility by $z + Q(t)/t^r$ in $\mathbb{R}[t, t^{-1}][z]$; and since t^r is a unit in $\mathbb{R}[t, t^{-1}]$, divisibility by $D = t^r(z + Q/t^r)$ follows, with a Laurent-polynomial quotient. This proves (2.2).

Divide (2.2) by D . Since S is a Laurent polynomial, it contains only finitely many powers of t . For all sufficiently large m , its t^m coefficient vanishes, and therefore

$$P_m(z) = [t^{m+rE}] \frac{A(t)}{D(t, z)}.$$

Factoring $A = t^\mu B$ gives (2.3). □

For all sufficiently large m , writing $B(t) = \sum_j b_j t^j$ and expanding (2.3) against (1.2) exhibits the reduced coefficient as a fixed finite linear combination of the denominator-only sequence,

$$(2.4) \quad P_m(z) = \sum_j b_j H_{m+rE-\mu-j}(z),$$

with coefficients independent of m .

Thus the proof reduces to one fixed univariate numerator. Throughout, let

$$(2.5) \quad F_M(z) = [t^M] \frac{B(t)}{Q(t) + zt^r}, \quad B \in \mathbb{R}[t], \quad B(0) \neq 0.$$

Lemma 2.3. *For all sufficiently large M ,*

$$(2.6) \quad \deg F_M = \left\lfloor \frac{M}{r} \right\rfloor.$$

Proof. The geometric expansion

$$(2.7) \quad \frac{B(t)}{Q(t) + zt^r} = \sum_{j \geq 0} (-z)^j t^{rj} \frac{B(t)}{Q(t)^{j+1}}$$

gives

$$(2.8) \quad F_M(z) = \sum_{0 \leq j \leq M/r} (-z)^j [t^{M-rj}] \frac{B(t)}{Q(t)^{j+1}}.$$

Hence $\deg F_M \leq \lfloor M/r \rfloor$.

Put

$$\ell = \left\lfloor \frac{M}{r} \right\rfloor, \quad s = M - r\ell, \quad 0 \leq s < r.$$

The coefficient of z^ℓ is

$$(2.9) \quad (-1)^\ell [t^s] \frac{B(t)}{Q(t)^{\ell+1}}.$$

Let

$$\lambda = -\frac{Q'(0)}{Q(0)} = \sum_{j=1}^n \frac{1}{x_j} > 0.$$

For each fixed s , expanding $Q(t)^{-(\ell+1)} = Q(0)^{-(\ell+1)} \prod_{j=1}^n (1 - t/x_j)^{-(\ell+1)}$ and extracting the coefficient of t^s shows that

$$Q(0)^{\ell+1} [t^s] \frac{B(t)}{Q(t)^{\ell+1}}$$

is a polynomial in ℓ of degree at most s , with leading coefficient $B(0)\lambda^s/s!$. Since $B(0)\lambda^s \neq 0$, it does not vanish identically and is therefore nonzero for all sufficiently large ℓ , so the upper bound on the degree is attained. □

Remark 2.4. The shift $M = m + rE - \mu$ in Lemma 2.2 is fixed. Thus

$$\deg P_m = \frac{m}{r} + O_{Q,r,N}(1)$$

eventually.

3. POLE GEOMETRY AND THE WEIGHTED PRINCIPAL AMPLITUDE

This section records the denominator facts needed in the weighted argument. They are established in [5], except for the case $(\deg Q, r) = (2, 1)$, a quadratic denominator excluded there because τ is then constant; that case is elementary and is recorded in Remark 3.2. The linear modulus gap (3.3) is derived below. Lemmas 3.3, 3.5 and 3.6 then group the nonprincipal roots by a fixed contour, treat an escaping root, and give the endpoint form and phase bound for the weighted principal amplitude W of (3.7); Sections 4 and 5 consume all three.

Let

$$g(t) = -\frac{Q(t)}{t^r}.$$

Following Forgács and Tran, let $t_a > 0$ be the smallest positive critical point of g , and set $b = +\infty$ when $r > 1$, while for $r = 1$ let $t_b < 0$ be the unique negative critical point of g [5, Lem. 5] and put $b = g(t_b)$. This is the endpoint convention of [5, Thm. 1], where both endpoints are read off the polynomial $t^{r-1}(rQ(t) - tQ'(t))$; its zero of order $r - 1$ at the origin is what makes the upper endpoint unbounded for $r > 1$. Set

$$(3.1) \quad a = g(t_a), \quad I_{Q,r} = (a, b).$$

Theorem 3.1 (Forgács–Tran denominator geometry). *There are real-analytic functions*

$$\tau : (0, \pi/r) \rightarrow (0, \infty), \quad z : (0, \pi/r) \rightarrow I_{Q,r},$$

with z strictly increasing, $z(\theta) \rightarrow a$ as $\theta \downarrow 0$ and $z(\theta) \rightarrow b$ as $\theta \uparrow \pi/r$, such that

$$(3.2) \quad t_{\pm}(\theta) = \tau(\theta)e^{\pm i\theta}$$

are zeros of $Q(t) + z(\theta)t^r$. They are the only denominator zeros in the closed disk $|t| \leq \tau(\theta)$.

On every compact subinterval of $(0, \pi/r)$, the remaining denominator zeros are separated from the principal pair by a uniform modulus ratio. Near each endpoint, write $\zeta_j(\theta) = t_j(\theta)/\tau(\theta)$ for the normalized denominator zeros. At $\theta \downarrow 0$, a smallest zero x_1 of Q of multiplicity $\rho \geq 2$ produces a cluster of ρ zeros tending to x_1 , whose $\rho - 2$ nonprincipal members satisfy a linear modulus gap

$$(3.3) \quad |\zeta_j(\theta)| \geq 1 + c\delta,$$

for some $c > 0$, δ denoting the angular distance to the endpoint, while every zero outside the cluster satisfies

$$(3.4) \quad |\zeta_j(\theta)| \geq 1 + c_*$$

for a fixed $c_ > 0$. When $\rho = 1$ the cluster is the principal pair alone and every nonprincipal zero satisfies (3.4). At $\theta \uparrow \pi/r$ with $r > 1$, the finite normalized cluster approaches the r th roots of -1 and its nonprincipal members satisfy (3.3), while every other normalized zero tends to infinity. At $\theta \uparrow \pi$ with $r = 1$, the principal pair coalesces at t_b to a zero of multiplicity exactly two, and every other normalized zero satisfies (3.4).*

Proof. The existence, strict monotonicity, and endpoint limits of τ and z are established in Lemmas 2–6 of [5]. The minimum-modulus assertion is [5, Props. 1–2]. Both exclude $(\deg Q, r) = (2, 1)$, where τ is constant; that case is Remark 3.2. Compact-uniform separation follows by continuity of the roots, compactness, and the strict pointwise modulus gap.

The endpoint statement is the quantitative content of the second and third cases in the proof of [5, Prop. 3]. At the lower endpoint, a smallest zero of multiplicity $\rho \geq 2$ gives normalized cluster roots

$$\zeta_j(\theta) = 1 + \frac{\cos(\pi/\rho) - \omega_j}{\sin(\pi/\rho)} \theta + O(\theta^2), \quad \omega_j = e^{(2j-1)\pi i/\rho},$$

so that

$$|\zeta_j(\theta)| = 1 + \frac{\cos(\pi/\rho) - \Re \omega_j}{\sin(\pi/\rho)} \theta + O(\theta^2);$$

the linear coefficient vanishes exactly for the two principal indices $\omega_j = e^{\pm i\pi/\rho}$ and is strictly positive otherwise. At the upper endpoint for $r > 1$, with $\eta = \pi/r - \theta$ and ω_j now the r th roots of -1 ,

$$|\zeta_j(\theta)| = 1 + \frac{\cos(\pi/r) - \Re \omega_j}{\sin(\pi/r)} \eta + O(\eta^2),$$

again with vanishing linear coefficient only for the principal pair $\omega_j = e^{\pm i\pi/r}$. Since the clusters are finite, the positive linear coefficients have a uniform positive minimum and the remainders are uniform over the cluster, giving (3.3). For $\rho = 2$ the cluster is the principal pair alone and (3.3) is vacuous.

The zeros outside the cluster keep a fixed gap. At the lower endpoint $\tau(\theta) \rightarrow t_a$ and $z(\theta) \rightarrow a$; among the zeros of $Q(t) + z(\theta)t^r$ that remain bounded, the normalized values converge to the zeros of the limit polynomial obtained at $(\tau, z) = (t_a, a)$, namely $\zeta = 1$ with multiplicity $\max\{\rho, 2\}$ together with the ratios t_j/t_a formed from the remaining bounded zeros t_j of $Q(t) + at^r$, each of modulus exceeding t_a . The minimum-modulus assertion is quantified over $\theta \in (0, \pi/r)$ and in the limit gives only $|t_j| \geq t_a$, so strictness is supplied directly. Let $t_e \neq 0$ and z_e be real with $Q(t_e) + z_e t_e^r = 0$. For $|t| = |t_e|$ with $t \neq t_e$ one has $\Re t < t_e$ if $t_e > 0$ and $\Re t > t_e$ if $t_e < 0$, so

$$|t - x_j|^2 - |t_e - x_j|^2 = 2x_j(t_e - \Re t) \quad (1 \leq j \leq n)$$

has the sign of t_e for every j ; hence $|Q(t)| \neq |Q(t_e)| = |z_e| |t_e|^r = |z_e t^r|$ and t is not a zero of $Q(t) + z_e t^r$ [5, Prop. 3]. At the lower endpoint this applies with $t_e = t_a$ and $z_e = a$. For $\rho \geq 2$, both Q and Q' vanish at x_1 , so x_1 is a critical point of g ; as $tQ'(t) - rQ(t) < 0$ on $(0, x_1)$, it is the smallest positive one. Hence $t_a = x_1$ and $a = g(x_1) = -Q(x_1)/x_1^r = 0$, so the limit polynomial is

$$Q(0)(1 - \zeta)^\rho \prod_{j>\rho} \left(1 - \frac{x_1 \zeta}{x_j}\right),$$

whose zeros off $\zeta = 1$ are the fixed ratios $x_j/x_1 > 1$. Continuity of the zeros in the coefficients gives (3.4) for any c_* below the least of these ratios diminished by one.

At the upper endpoint for $r = 1$ the comparison above applies with $t_e = t_b < 0$ and $z_e = b$, the sign in the display reversing with t_e . A multiplicity above two at t_b would force $Q''(t_b) = 0$, whereas the zeros of Q'' are positive; the collision is therefore exactly double, and $\deg Q \geq 2 > r$ leaves no escaping zero, the leading t -coefficient of $Q(t) + zt$ being the nonzero constant $q_{\deg Q}$, so the remaining $\deg Q - 2$ zeros satisfy (3.4) by continuity of the zeros in the coefficients [5, Prop. 3]. The zeros that escape to infinity are treated in Lemma 3.5 and in the proof of Theorem 4.1. The upper endpoint for $r > 1$ is the third case of the same proof, where the normalized zeros outside the finite cluster tend to infinity. $\square$

Remark 3.2. For $\deg Q = 2$ and $r = 1$, write $Q(t) = q_0 + q_1 t + q_2 t^2$, so that $q_0, q_2 > 0$ and $q_1 < 0$. The two zeros of $Q(t) + zt$ have product q_0/q_2 , so the conjugate pair (3.2) has $\tau = \sqrt{q_0/q_2}$, constant in z , and

$$z(\theta) = -q_1 - 2\sqrt{q_0 q_2} \cos \theta \quad (0 < \theta < \pi),$$

which is real-analytic and strictly increasing, with $t_a = \sqrt{q_0/q_2} = -t_b$ and $I_{Q,r} = (-q_1 - 2\sqrt{q_0 q_2}, -q_1 + 2\sqrt{q_0 q_2})$. The pair exhausts the zeros of the denominator, so the minimum-modulus assertion is immediate, the collision at either endpoint is exactly double, and (3.3) and (3.4) are vacuous.

As z is a strictly monotone bijection onto $I_{Q,r}$, its inverse is written $\theta(z)$, and $\tau(z)$ abbreviates $\tau(\theta(z))$.

The coefficient asymptotics are easiest away from multiple denominator roots. If $t_1, \dots, t_d$ are the simple roots of $Q(t) + zt^r$, then for every $M > \deg B - d$

$$(3.5) \quad F_M(z) = - \sum_{j=1}^d \frac{B(t_j)}{t_j^{M+1} (Q'(t_j) + r z t_j^{r-1})},$$

the polynomial part of $B/(Q + zt^r)$, of t -degree $\deg B - d$, contributing nothing to $[t^M]$ in this range. At a nonprincipal collision, the individual terms may be singular even though their sum is regular. Such roots are always grouped by a fixed contour.

Lemma 3.3. *Let K be a compact parameter interval in $I_{Q,r} \cup \{a\}$, or in $I_{Q,r} \cup \{a, b\}$ when $r = 1$, and let Γ be a contour enclosing a complete cluster of denominator roots and no other roots for every z in a neighborhood of K . Then the contribution of that cluster to $F_M(z)$ is holomorphic in z and equals*

$$(3.6) \quad -\frac{1}{2\pi i} \int_{\Gamma} \frac{B(t)}{t^{M+1}(Q(t) + zt^r)} dt.$$

If the contour may be chosen in $|t| \geq R_0 > \tau(z)$, then after multiplication by $\tau(z)^{M+1}$ the contribution is

$$O_K(\sigma^M)$$

for some fixed $\sigma < 1$.

Proof. The contour integral is invariant under the internal splitting or collision of the roots and is holomorphic by differentiation under the integral sign. On Γ , where $|t| \geq R_0$, $|B(t)|$ is bounded above and $|Q(t) + zt^r|$ is bounded below, so the contour estimate bounds the contribution by $C_K R_0^{-M}$ with C_K depending on the length of Γ , the upper bound for B , and the lower bound for the denominator; multiplying by $\tau(z)^{M+1}$ and using $\tau(z) \leq \sigma R_0$ with $\sigma < 1$ uniform on K gives $O_K(\sigma^M)$. Only $\sup_K \tau < R_0$ is used, which holds at either finite endpoint, where $\tau(a) = t_a$ and, for $r = 1$, $\tau(b) = |t_b|$ are finite. A finite cover of the compact interval supplies finitely many such contours. $\square$

Remark 3.4. That the contour of Lemma 3.3 encloses the same cluster for every z in a neighborhood of K follows, for the principal cluster, from the minimum-modulus theorem of Forgács and Tran, Theorem 3.1; for a nonprincipal cluster it follows from continuity of the denominator zeros in z together with the separation of that cluster from the others at each parameter of K .

For $0 < \theta < \pi/r$, define the weighted principal amplitude

$$(3.7) \quad W(\theta) = -\frac{B(t_+(\theta))}{Q'(t_+(\theta)) + rz(\theta)t_+(\theta)^{r-1}}.$$

The denominator in (3.7) equals $-t_+(\theta)^r g'(t_+(\theta))$, since $z(\theta) = -Q(t_+)/t_+^r$ and $tQ'(t) - rQ(t) = -t^{r+1}g'(t)$. A double principal root would make $t_+(\theta)$ a critical point of g , that is, would force $t_+Q'(t_+) = rQ(t_+)$. This cannot occur. The zeros of Q are positive, so Q does not vanish off the real axis, and with $x_1, \dots, x_n$ as in (1.1), listed with multiplicity,

$$\Im \frac{tQ'(t)}{Q(t)} = \Im \sum_{j=1}^n \frac{t}{t - x_j} = -\Im t \sum_{j=1}^n \frac{x_j}{|t - x_j|^2} < 0 \quad (\Im t > 0),$$

whereas r is real. Since $t_+(\theta) = \tau(\theta)e^{i\theta}$ with $0 < \theta < \pi/r \leq \pi$ lies in the open upper half-plane, $g'(t_+(\theta)) \neq 0$ for every such θ , and W is finite and real-analytic on $(0, \pi/r)$. Combining the two conjugate terms in (3.5) gives

$$(3.8) \quad \tau(\theta)^{M+1} F_M(z(\theta)) = 2\Re \left(W(\theta) e^{-i(M+1)\theta} \right) + R_M(\theta),$$

where R_M is the grouped normalized contribution of every nonprincipal root. Figure 1(A) shows the decomposition for one numerator.

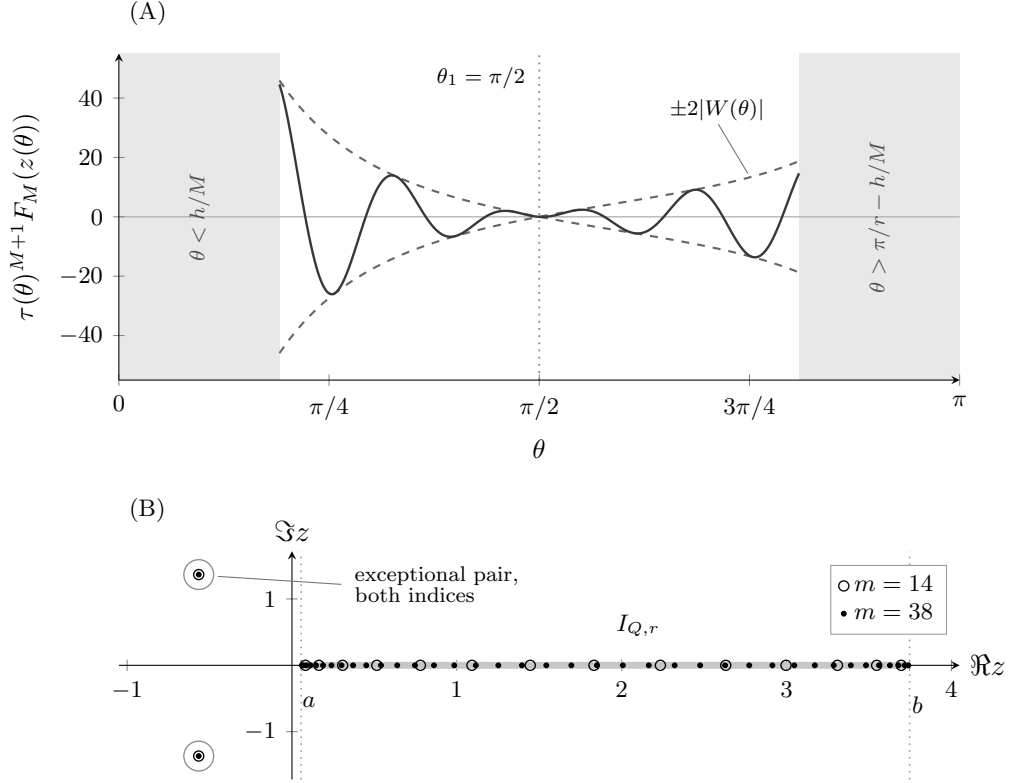


FIGURE 1. One denominator, two numerators. Throughout $Q(t) = (1 - t)(1 - t/2)(1 - t/4)$ and $r = 1$. The upper endpoint is finite because $r = 1$, and the smallest zero of Q is simple, so $a > 0$; thus $I_{Q,r} = (a, b) = (0.0559\dots, 3.7466\dots)$. **(A)** The decomposition (3.8) at $M = 14$ for $N(t, z) = B(t) = 7t^2 + 8$, which Lemma 2.2 leaves unshifted, with the principal-pair envelope $\pm 2|W(\theta)|$ of (3.7) dashed. Consecutive phase points, those at which $(M + 1)\theta - \arg W(\theta)$ is an integer multiple of π , enclose distinct zeros in $I_{Q,r}$ (Proposition 5.1). Since $t_+(\pi/2) = i\sqrt{8/7}$ and $z(\pi/2) = 45/28$, one has $B(t_+(\pi/2)) = 0$, so W has an amplitude zero at $\theta_1 = \pi/2$ and the envelope pinches there; the neighborhood of θ_1 deleted in (4.1) is exponentially small in M . Numerically, the dominance inequality (4.2) fails only within 10^{-11} of θ_1 at this M , too narrow to draw. The shaded endpoint windows of (4.3) are drawn wider than h requires, which keeps on scale an envelope that grows without bound at either endpoint. **(B)** Zeros of P_m for the numerator $N(t, z) = (1 + z + z^2) + t(2 - z)$, which is not of the form $R(z)S(t)$: its exceptional zeros are created by the initial conditions rather than factored out of it as in Proposition 6.3. At each displayed index, $\deg P_m = m + 2$, with m simple zeros in $I_{Q,r}$ and exactly two elsewhere, a conjugate pair at $-0.5655\dots \pm 1.3674\dots i$ agreeing to thirteen decimals at the two indices. The bulk grows with m ; the defect does not.

Lemma 3.5. *The d -term formula (3.5) presumes d finite roots, and a root escapes to infinity exactly where the leading t -coefficient of $Q(t) + zt^r$ vanishes. When $\deg Q \neq r$ no such parameter lies in $I_{Q,r}$; when $\deg Q = r = d$ the leading coefficient vanishes at exactly one parameter, $z_* = -q_d$ with q_d the leading coefficient of Q . If $z_* \notin \overline{I_{Q,r}}$, it does not occur on the parameter interval. If $z_* \in \overline{I_{Q,r}}$, then for $M > \deg B - d + 1$ the escaping root's contribution to F_M extends holomorphically across z_* , vanishing there, and on a sufficiently small relative neighborhood $U \subset I_{Q,r}$ of z_* its normalization by $\tau(z)^{M+1}$ is $O(\sigma^M)$ for some $\sigma < 1$. The escaping root therefore requires no separate treatment once $M > \deg B - d + 1$. If $z(\theta) \rightarrow 0$ at the lower endpoint and $\deg Q < r$, then $r - \deg Q$ roots escape as $\theta \downarrow 0$; their grouped endpoint contribution is treated in the proof of Theorem 4.1.*

Proof. When $\deg Q < r$ the leading coefficient is z , which vanishes only at $z = 0 \notin I_{Q,r}$; when $\deg Q > r$ it is the nonzero constant $q_{\deg Q}$; and when $\deg Q = r = d$ it is $q_d + z$, vanishing exactly at $z_* = -q_d$.

In the reciprocal coordinate $u = 1/t$ put

$$\widehat{D}(u, z) = u^d Q(1/u) + z, \quad \widehat{B}(u) = u^{\deg B} B(1/u).$$

Then $\widehat{D}(0, z_*) = 0$ and $\partial_u \widehat{D}(0, z_*) = q_{d-1} \neq 0$ (the zeros of Q are positive), so a unique analytic root $u_\infty(z)$ with $u_\infty(z_*) = 0$ describes the escaping root, and its contribution to $F_M(z)$ is

$$(3.9) \quad \frac{u_\infty(z)^{M+d-1-\deg B} \widehat{B}(u_\infty(z))}{\partial_u \widehat{D}(u_\infty(z), z)}.$$

The condition $M > \deg B - d + 1$ makes the exponent $M + d - 1 - \deg B$ positive, so (3.9) extends holomorphically across z_* and vanishes there. Assume $z_* \in \overline{I_{Q,r}}$ and restrict z to a sufficiently small relative neighborhood $U \subset I_{Q,r}$ of z_* . Being positive and continuous, τ is bounded above and bounded away from zero on U , using its finite endpoint limit when z_* is an endpoint. Since $u_\infty(z_*) = 0$, shrinking U gives

$$\sup_{z \in U} \tau(z) |u_\infty(z)| \leq \sigma < 1,$$

so after multiplication by $\tau(z)^{M+1}$ the contribution is $O(\sigma^M)$. $\square$

Lemma 3.6. *The function W has only finitely many zeros in $(0, \pi/r)$. If these are $\theta_1, \dots, \theta_J$, with multiplicities $\nu_1, \dots, \nu_J$, then near θ_j ,*

$$(3.10) \quad W(\theta) = (\theta - \theta_j)^{\nu_j} U_j(\theta), \quad U_j(\theta_j) \neq 0.$$

Near each endpoint,

$$(3.11) \quad W(\theta) = \delta^p V(\delta),$$

where δ is the angular distance to the endpoint, p is a fixed integer, and V is real-analytic and nonvanishing at $\delta = 0$; at a finite endpoint, where the principal pair coalesces to a zero of the limiting denominator of multiplicity k and B vanishes to order ν , one has $p = \nu - (k - 1)$, with $k = \max\{\rho, 2\}$ at the lower endpoint, ρ denoting the multiplicity of the smallest zero of Q , and $k = 2$ at the upper endpoint when $r = 1$; at the upper endpoint when $r > 1$ one has $p = 1$. The integer p may be negative, in which case $|W| \rightarrow \infty$. Consequently, on each component of $(0, \pi/r) \setminus \{\theta_1, \dots, \theta_J\}$ one can choose a continuous branch $\psi = \arg W$ with

$$(3.12) \quad |\psi'(\theta)| \leq \kappa$$

for a constant κ depending only on Q, r, B ; since ψ is selected on each component separately, it is the phase variations of these branches that sum to a bound depending only on the same data.

Proof. Since $\arg t_+(\theta) = \theta$, the map $\theta \mapsto t_+(\theta)$ is injective, so $B(t_+(\theta))$ vanishes at no more than $\deg B$ distinct points of $(0, \pi/r)$, at which the denominator in (3.7) does not vanish; let ν_j be their multiplicities as zeros of the analytic composition $B(t_+(\theta))$. This gives the first assertion and (3.10).

Along the real parameter θ , the factor $(\theta - \theta_j)^{\nu_j}$ has constant argument on each side of θ_j , so the real term $\nu_j/(\theta - \theta_j)$ in W'/W does not contribute to $\Im(W'/W) = \psi'$, so ψ' remains bounded on each component adjacent to θ_j , with finite one-sided limits at the deleted point.

By Theorem 3.1, τ and $t_{\pm}$ are real-analytic on $(0, \pi/r)$. Consider a finite endpoint, so $\theta \downarrow 0$ with $\delta = \theta$, or, when $r = 1$, $\theta \uparrow \pi$ with $\delta = \pi - \theta$; write t_e and z_e for the limits of $t_+(\theta)$ and $z(\theta)$ there. Then t_e is real and nonzero, being $t_a > 0$ at the lower endpoint and $t_b < 0$ at the upper one, and the principal pair coalesces at t_e , a zero of the limiting denominator $Q(t) + z_e t^r$ of multiplicity $k \geq 2$. Write $Q(t) + z_e t^r = (t - t_e)^k G(t)$ with $G(t_e) \neq 0$, so that a zero t of $Q(t) + z t^r$ near t_e satisfies

$$(t - t_e)^k = -(z - z_e) \frac{t^r}{G(t)}.$$

On the interior side $z - z_e > 0$ at the lower endpoint and $z - z_e < 0$ at the upper one, so that $z - z_e = \pm y^k$ with $y \geq 0$ real and the sign fixed by the endpoint. Since $t_e \neq 0$ and $G(t_e) \neq 0$, the function $\mp t^r/G(t)$ is analytic and nonvanishing near t_e ; fix an analytic branch Λ of $(\mp t^r/G(t))^{1/k}$ there. Each branch of the cluster satisfies $t - t_e = \omega y \Lambda(t)$ with $\omega^k = 1$, and the implicit function theorem gives a unique $t(y)$, analytic in the real parameter y , with

$$t(y) = t_e + \gamma y + O(y^2), \quad \gamma = \omega \Lambda(t_e) \neq 0.$$

The k values $\omega \Lambda(t_e)$ are distinct, so γ determines the branch. For real y the equation $Q(t) + (z_e \pm y^k)t^r = 0$ has real coefficients, so $y \mapsto \overline{t(y)}$ solves it near t_e and is again one of the k branches, with leading coefficient $\bar{\gamma}$; a branch whose γ is real is therefore its own conjugate, hence real. The principal pair is nonreal for $0 < \theta < \pi/r$, so its leading coefficient γ_0 has $\Im \gamma_0 \neq 0$. As t_e is real and nonzero, $\delta = \pm \Im \log(t_+/t_e)$ in the branch of the logarithm analytic at 1, so δ is real-analytic in y with

$$\left| \frac{d\delta}{dy}(0) \right| = \left| \Im \frac{\gamma_0}{t_e} \right| \neq 0,$$

and inverting, t_+ is real-analytic in δ up to the endpoint:

$$(3.13) \quad t_+(\theta) = t_e + \gamma_+ \delta + O(\delta^2), \quad \gamma_+ \neq 0, \quad \frac{dt_+}{d\delta} = \gamma_+ + O(\delta).$$

Since t_+ is a denominator zero, $z(\theta) = -Q(t_+)/t_+^r$, so the derivative factor in (3.7) is

$$Q'(t_+) + rz(\theta)t_+^{r-1} = Q'(t_+) - \frac{rQ(t_+)}{t_+} = -t_+^r g'(t_+).$$

At either finite endpoint B and $t^r g'(t) = (rQ(t) - tQ'(t))/t$ are analytic in a neighborhood of $t_e \neq 0$, vanishing there to finite orders ν and $k-1$: from $Q(t) + z_e t^r = t^r(z_e - g(t))$ the function $z_e - g$ vanishes at t_e to order k , and $t^r g' = -t^r(z_e - g)'$. By Theorem 3.1, $k = \max\{\rho, 2\}$ at the lower endpoint and $k = 2$ at the finite upper endpoint. Writing $w = t_+ - t_e$, which by (3.13) is $\gamma_+ \delta(1 + O(\delta))$,

$$W(\theta) = \frac{B(t_+)}{t_+^r g'(t_+)} = \frac{c_B w^\nu (1 + O(w))}{c_g w^{k-1} (1 + O(w))}, \quad c_B, c_g \neq 0,$$

so (3.11) holds with $p = \nu - (k-1)$ and V real-analytic and nonvanishing at $\delta = 0$.

At the upper endpoint when $r > 1$, set $\eta = \pi/r - \theta$. Here $t_+(\theta) \rightarrow 0$, so $Q'(t_+) - rQ(t_+)/t_+ = -rQ(0)t_+^{-1}(1 + o(1))$ has a simple pole rather than a finite-order zero; instead write

$$W(\theta) = -\frac{t_+(\theta) B(t_+(\theta))}{t_+(\theta) Q'(t_+(\theta)) - rQ(t_+(\theta))}.$$

The denominator is analytic in t_+ and tends to $-rQ(0) \neq 0$. Here the collision is at $t = 0$ as $z \rightarrow +\infty$: from $t^r = -Q(t)/z$ one gets $t = \omega \varsigma Q(t)^{1/r}$ with $\varsigma = z^{-1/r}$ and $\omega^r = -1$, so $t(\varsigma)$ is analytic with $\frac{dt}{d\varsigma}(0) = \omega Q(0)^{1/r} \neq 0$, while $\eta = -\frac{1}{r} \arg Q(t_+)$ has $\frac{d\eta}{d\varsigma}(0) = \frac{\lambda}{r} Q(0)^{1/r} \sin(\pi/r) > 0$ for $\lambda = -Q'(0)/Q(0) > 0$. Hence $t_+(\theta) = \eta T(\eta)$ with T real-analytic and $T(0) = r e^{i\pi/r} / (\lambda \sin(\pi/r)) \neq 0$. Since $B(0) \neq 0$, it follows that $W(\theta) = \eta V(\eta)$ with V real-analytic and $V(0) \neq 0$, so (3.11) holds with $p = 1$.

For the phase, differentiate the two factors separately in δ . Both are analytic at the endpoint limit, vanishing to orders ν and $k-1$, so by (3.13), which gives $d \log w / d\delta = \delta^{-1}(1 + O(\delta))$,

$$\frac{d}{d\delta} \log W = (\nu - (k-1)) \frac{d}{d\delta} \log w + O(1) = \frac{p}{\delta} + O(1).$$

Since δ is θ or $\pi - \theta$ according to the endpoint, differentiating in θ instead only changes the sign, and the singular part $\pm p/\delta$ is real either way, so it drops out of $\Im(W'/W) = \psi'$, exactly as the interior term $\nu_j/(\theta - \theta_j)$ does above. The same holds at the upper endpoint for $r > 1$, where (3.11) holds with $p = 1$. Hence ψ' is bounded up to both endpoints, which gives (3.12) on each component with κ determined by Q, r, B ; the components being disjoint subintervals of $(0, \pi/r)$, the phase variations sum to at most $\kappa\pi/r$. $\square$

4. WEIGHTED PRINCIPAL-PAIR DOMINANCE

This section extends the estimates in the proof of [5, Prop. 3] from the constant residue weight to the fixed polynomial weight B .

Theorem 4.1 (Weighted principal-pair dominance). *There exist constants $h > 0$, M_0 , and $c > 0$, depending only on Q, r, B , with the following property. For every $M \geq M_0$, let*

$$(4.1) \quad \Theta_{j,M} = \left\{ \theta : |\theta - \theta_j| < e^{-cM/\nu_j} \right\}$$

for the zeros in Lemma 3.6. Then

$$(4.2) \quad |R_M(\theta)| \leq \frac{1}{2} |W(\theta)|$$

whenever

$$(4.3) \quad \frac{h}{M} \leq \theta \leq \frac{\pi}{r} - \frac{h}{M}, \quad \theta \notin \bigcup_{j=1}^J \Theta_{j,M}.$$

The total length of the deleted amplitude intervals is exponentially small.

Proof. We divide the parameter interval into the same three regions as Forgács and Tran.

Compact interior. Fix $\varepsilon \in (0, \pi/(2r))$ small enough that every zero θ_j of Lemma 3.6 lies in $[\varepsilon, \pi/r - \varepsilon]$ and that the endpoint expansions of Theorem 3.1 hold on $(0, \varepsilon]$ and on $[\pi/r - \varepsilon, \pi/r)$; the three regions below then cover $(0, \pi/r)$, and the deleted windows $\Theta_{j,M}$ all lie in the compact interior. Since the θ_j are finite in number and independent of M , such an ε depends only on Q, r, B , as then do the constants below. On $[\varepsilon, \pi/r - \varepsilon]$, the minimum-modulus gap in Theorem 3.1 is compact-uniform. By Lemma 3.3 (with the escaping root at a degree-drop parameter handled by Lemma 3.5),

$$(4.4) \quad R_M(\theta) = O(\sigma^M)$$

for some $\sigma < 1$. Near an amplitude zero θ_j ,

$$|W(\theta)| \asymp |\theta - \theta_j|^{\nu_j}.$$

Choose $c \in (0, -\log \sigma)$, so that $e^{-cM} \gg \sigma^M$. Outside $\Theta_{j,M}$, we then have $|W| \gtrsim c' e^{-cM}$, with $c' > 0$ the constant implicit in the comparison above, and hence $|R_M| = o(|W|)$, the fixed constant c' being absorbed by enlarging M_0 . Away from the amplitude zeros, $|W|$ is bounded below and the conclusion is immediate.

The lower endpoint, $\theta \downarrow 0$. When the smallest zero of Q is simple, Theorem 3.1 gives a cluster consisting of the principal pair alone and the fixed gap (3.4) for every nonprincipal zero, so the nonprincipal remainder is $O(\sigma^M)$ as in the compact interior; near the endpoint $|W(\theta)| \asymp \theta^p$ by Lemma 3.6, so on $\theta \geq h/M$ one has $\sigma^M = o(|W(\theta)|)$ and (4.2) follows. It remains to consider a repeated smallest zero x_1 of Q , of multiplicity $\rho > 1$. The endpoint analysis of [5, Prop. 3] produces a finite cluster of actual roots $t_j(\theta) \rightarrow x_1$, with

$$(4.5) \quad t_j(\theta) = x_1 - \frac{x_1 \omega_j}{\sin(\pi/\rho)} \theta + O(\theta^2)$$

for $\omega_j = e^{(2j-1)\pi i/\rho}$ as in the proof of Theorem 3.1, by the expansion there together with $\tau(\theta) = x_1(1 - \cot(\pi/\rho)\theta) + O(\theta^2)$; each coefficient has modulus $x_1/\sin(\pi/\rho)$. The normalized values $\zeta_j = t_j/\tau$ satisfy $|\zeta_{\pm}(\theta)| = 1$ for the two principal roots and

$$(4.6) \quad |\zeta_j(\theta)| \geq 1 + c_0\theta$$

for every other cluster root, for some $c_0 > 0$. The numerator and derivative weights follow from (4.5). If B vanishes at x_1 to order ν , then

$$(4.7) \quad B(t_j(\theta)) = c_j\theta^\nu(1 + O(\theta)), \quad c_j \neq 0,$$

the same order ν for every member, so the ratios $B(t_j)/B(t_+)$ have absolute values bounded above and bounded away from zero. For the derivative factor, t_j is a denominator root, so $z(\theta) = -Q(t_j)/t_j^r$ and

$$(4.8) \quad Q'(t_j) + rz(\theta)t_j^{r-1} = Q'(t_j) - \frac{rQ(t_j)}{t_j}.$$

As Q vanishes to order ρ at x_1 , the first term is of order $\theta^{\rho-1}$ and the second of order θ^ρ , so the derivative factor has leading order $\theta^{\rho-1}$ across the cluster and its ratios have absolute values bounded above and bounded away from zero. Each normalized nonprincipal term is therefore an $O(1)$ multiple of $|\zeta_j(\theta)|^{-(M+1)}$ times the principal amplitude, and by (4.6) at most

$$Ce^{-c_0M\theta}$$

times it. When $\theta \geq h/M$, this is at most Ce^{-c_0h} . Choosing h sufficiently large makes the total contribution of the finite nonprincipal cluster less than $|W|/4$. The zeros outside the cluster keep the fixed modulus gap (3.4) of Theorem 3.1, so contribute $o(|W|)$ as in the compact interior; for $\rho = 2$, where the cluster is the principal pair alone, this accounts for every nonprincipal zero. Since $\rho > 1$, the proof of Theorem 3.1 gives $t_a = x_1$ and $a = 0$, so that $z(\theta) \rightarrow 0$. If in addition $\deg Q < r$, the leading coefficient of $Q(t) + z(\theta)t^r$ is $z(\theta)$ (Lemma 3.5) and $r - \deg Q$ roots escape to infinity as $\theta \downarrow 0$. Fix R_0 exceeding every zero of Q . For small θ , Rouché's theorem on $|t| = R_0$ separates the $\deg Q$ finite roots from the escaping ones, whose grouped contribution equals

$$\frac{1}{2\pi i} \int_{|t|=R_0} \frac{B(t)}{t^{M+1}(Q(t) + z(\theta)t^r)} dt$$

since for $M > \deg B - r$ the integrand is $O(t^{-2})$ as $t \rightarrow \infty$, so that its residue at infinity vanishes. On $|t| = R_0$ one has $|Q(t) + z(\theta)t^r| \geq c_{R_0} > 0$ for small θ , so after multiplication by $\tau(\theta)^{M+1}$ this contribution is $O((\tau(\theta)/R_0)^M)$; since $\tau(\theta) \rightarrow x_1 < R_0$ and $|W(\theta)| \asymp \theta^p$ with p fixed, it is $o(|W(\theta)|)$ on $\theta \geq h/M$, whatever the sign of p .

The upper endpoint, $\theta \uparrow \pi/r$. When $r = 1$, the endpoint is a finite collision and the nonprincipal remainder is $O(\sigma^M)$ as in the compact interior, the nonprincipal contributions being grouped by Lemma 3.3, whose parameter interval may include b in this case. With $\eta = \pi - \theta$, one has $|W(\theta)| \asymp \eta^p$ by Lemma 3.6, so on $\eta \geq h/M$ one has $\sigma^M = o(|W(\theta)|)$ and (4.2) follows. A possible zero of B at the collision point contributes the same algebraic order to the two principal residues, while every nonprincipal cluster has a fixed modulus gap.

Assume $r > 1$ and set

$$\eta = \pi/r - \theta.$$

By Theorem 3.1, the finite normalized endpoint cluster approaches the r th roots of -1 , the principal roots approaching $e^{\pm i\pi/r}$, every other finite-cluster root satisfies (3.3) at $\delta = \eta$ with a constant $c_1 > 0$, and all remaining normalized roots tend to infinity. Here $\tau(\theta) \rightarrow 0$; by (4.8) and $t_j = \tau\zeta_j$,

$$Q'(t_j) + rz(\theta)t_j^{r-1} = -\frac{rQ(0)}{\tau\zeta_j}(1 + o(1)),$$

so the derivative factors of the small-root cluster differ only by the bounded ratios ζ_+/ζ_j , the common factor τ^{-1} cancelling against the principal one. Since $B(0) \neq 0$, the numerator weights

of the small-root cluster have absolute values bounded above and bounded away from zero. Hence the finite small-root cluster contributes at most

$$Ce^{-c_1 M \eta}$$

times the principal amplitude, at most $Ce^{-c_1 h}$ for $\eta \geq h/M$. If $\deg Q \leq r$, there are no remaining roots. If $\deg Q > r$, the remaining $\deg Q - r$ roots escape to infinity as $z(\theta) \rightarrow +\infty$. Fix $R_0 > 0$. For all sufficiently small η , Rouché's theorem on $|t| = R_0$ shows that the r roots of the small-root cluster lie inside the circle and every remaining root lies outside it. For all sufficiently large M , the grouped contribution C_{out} of the outer roots is therefore

$$C_{\text{out}}(\theta) = \frac{1}{2\pi i} \int_{|t|=R_0} \frac{B(t)}{t^{M+1}(Q(t) + z(\theta)t^r)} dt.$$

Indeed, for $M > \deg B - d$ the integrand is $O(t^{-2})$ as $t \rightarrow \infty$, so its residue at infinity vanishes; the contour encloses the pole at $t = 0$ and exactly the small-root cluster, and the residue theorem then identifies the integral with the complementary root contribution.

On $|t| = R_0$, for sufficiently small η ,

$$|Q(t) + z(\theta)t^r| \geq \frac{1}{2}z(\theta)R_0^r.$$

Consequently,

$$|\tau(\theta)^{M+1}C_{\text{out}}(\theta)| \leq C\tau(\theta)^{M+1}z(\theta)^{-1}R_0^{-M-r}.$$

Since

$$z(\theta)^{-1} = O(\tau(\theta)^r) \quad \text{and} \quad |W(\theta)| \asymp \tau(\theta),$$

this is

$$O\left(\left(\frac{\tau(\theta)}{R_0}\right)^{M+r}\right)|W(\theta)|,$$

uniformly negligible in the upper-endpoint region. Increasing h therefore gives (4.2).

The estimates are uniform on the overlaps of the three regions. The total length of the $\Theta_{j,M}$ is exponentially small because their number and multiplicities are fixed. $\square$

Remark 4.2. The proof does not require the exact sign of the full weighted endpoint cluster. Outside angular windows of width h/M , the linear endpoint modulus gap becomes the fixed exponential suppression $e^{-c_0 h}$ at the lower endpoint and $e^{-c_1 h}$ at the upper.

5. PROOF OF THE FIXED-NUMERATOR THEOREM

Proposition 5.1 treats the reduced sequence F_M of (2.5); the reduction of Lemma 2.2 then gives Theorem 1.1. Proposition 5.2 closes the excluded case $\deg Q = r = 1$.

Proposition 5.1. *Let $B \in \mathbb{R}[t]$ satisfy $B(0) \neq 0$, and define F_M by (2.5). There is a constant $C_B = C(Q, r, B)$ such that, for all sufficiently large M ,*

$$(5.1) \quad \#\{x \in I_{Q,r} : F_M(x) = 0\} \geq \left\lfloor \frac{M}{r} \right\rfloor - C_B,$$

where the zeros on the left are counted distinctly. Consequently F_M has at most C_B zeros outside $I_{Q,r}$, counted with multiplicity.

Proof. Let h be chosen as in Theorem 4.1, and define the retained set

$$(5.2) \quad \Omega_M = \left[\frac{h}{M}, \frac{\pi}{r} - \frac{h}{M} \right] \setminus \bigcup_{j=1}^J \Theta_{j,M}.$$

It has a number of connected components bounded independently of M , and

$$(5.3) \quad |\Omega_M| = \frac{\pi}{r} - \frac{2h}{M} + o(M^{-1}).$$

On each component choose a continuous branch $\psi(\theta) = \arg W(\theta)$ and put

$$(5.4) \quad \Phi_M(\theta) = (M+1)\theta - \psi(\theta).$$

By Lemma 3.6,

$$\Phi'_M(\theta) = M + 1 - \psi'(\theta) > 0$$

for all sufficiently large M . Thus Φ_M is strictly increasing on each component.

At a phase point ϑ_k satisfying

$$\Phi_M(\vartheta_k) = k\pi,$$

the principal term in (3.8) is

$$2(-1)^k |W(\vartheta_k)|.$$

The dominance estimate (4.2) shows that the complete value $F_M(z(\vartheta_k))$ has the same sign. Consecutive phase points in one component therefore enclose a zero of $F_M(z(\theta))$. Since $z(\theta)$ is strictly monotone, all zeros obtained in this way are distinct points of $I_{Q,r}$.

On a component of Ω_M with endpoints $\theta^- < \theta^+$, the number of enclosed zeros is at least

$$\frac{\Phi_M(\theta^+) - \Phi_M(\theta^-)}{\pi} - 2.$$

Summing over the bounded number of components gives

$$\begin{aligned} \#\{x \in I_{Q,r} : F_M(x) = 0\} &\geq \frac{M+1}{\pi} |\Omega_M| - \frac{1}{\pi} \text{Var } \psi - O(1) \\ (5.5) \qquad \qquad \qquad &= \frac{M}{r} - O(1), \end{aligned}$$

where $\text{Var } \psi$ is the summed phase variation of the branches over the components of Ω_M , bounded by Lemma 3.6. This proves (5.1).

By Lemma 2.3,

$$\deg F_M = \left\lfloor \frac{M}{r} \right\rfloor$$

for all sufficiently large M . A polynomial of degree $\lfloor M/r \rfloor$ with at least $\lfloor M/r \rfloor - C_B$ distinct zeros in $I_{Q,r}$ has at most C_B remaining zeros, counted with multiplicity. This proves the last assertion. $\square$

Proof of Theorem 1.1. Apply Lemma 2.2. For all sufficiently large m ,

$$P_m(z) = F_{m+rE-\mu}(z)$$

for a fixed $B \in \mathbb{R}[t]$ satisfying $B(0) \neq 0$. The shifted index tends to infinity with m , so Proposition 5.1 gives a uniform bound for all sufficiently large m . In particular, these P_m are nonzero and $\deg P_m = \deg F_{m+rE-\mu} = \lfloor (m+rE-\mu)/r \rfloor$ by Lemma 2.3, so the distinct-zero count (5.1) reads $\#\{x \in I_{Q,r} : P_m(x) = 0\} \geq \deg P_m - C$. Enlarging the constant to cover the finitely many initial nonzero polynomials proves the theorem. $\square$

Proposition 5.2. *Suppose $\deg Q = r = 1$. Then the first conclusion of Theorem 1.1 holds: there is a C such that every nonzero P_m has at most C zeros outside $(0, \infty)$, counted with multiplicity. Its second conclusion is vacuous here, $I_{Q,r}$ being undefined for $\deg Q = r = 1$.*

Proof. Write

$$Q(t) = q_0 + q_1 t, \quad q_0 > 0, \quad q_1 < 0.$$

The properness condition $\deg_t N < 1$ forces $N(t, z) = R(z)$. Hence

$$\begin{aligned} \frac{R(z)}{q_0 + (q_1 + z)t} &= \sum_{m \geq 0} \frac{(-1)^m R(z)(q_1 + z)^m}{q_0^{m+1}} t^m, \\ P_m(z) &= \frac{(-1)^m}{q_0^{m+1}} R(z)(z + q_1)^m. \end{aligned}$$

The moving zero $-q_1$ is positive, and every exceptional zero belongs to the fixed polynomial R . $\square$

6. CONSEQUENCES AND SHARPNESS

After normalization by degree, the zeros of every fixed numerator converge weakly to a common limiting measure determined by the denominator. For $P \in \mathbb{R}[z]$ of positive degree k with zeros $\xi_1, \dots, \xi_k$ repeated by multiplicity, write $\mu[P] = \frac{1}{k} \sum_{i=1}^k \delta_{\xi_i}$ for its normalized zero-counting measure.

Proposition 6.1 (Equidistribution of the zero bulk). *Let $z: (0, \pi/r) \rightarrow I_{Q,r}$ be the parameterization of Theorem 3.1, and let μ_∞ be the pushforward under z of the uniform measure $\frac{r}{\pi} d\theta$ on $(0, \pi/r)$, a Borel probability measure on $\overline{I_{Q,r}}$ independent of N . Then, along the coefficient polynomials of positive degree,*

$$\int f d\mu[P_m] \longrightarrow \int f d\mu_\infty \quad \text{for every bounded continuous } f: \mathbb{C} \rightarrow \mathbb{R}.$$

Every fixed numerator therefore yields the same limiting zero distribution as the denominator-only sequence H_m .

Proof. By Lemma 2.2 it suffices to treat $\mu[F_M]$ as $M \rightarrow \infty$. Fix $0 < \alpha < \beta < \pi/r$ and let $Z_M[\alpha, \beta]$ be the number of zeros of F_M in $z([\alpha, \beta])$, counted with multiplicity.

Repeat the phase count of Proposition 5.1 on $[\alpha, \beta]$. After deleting the amplitude windows $\Theta_{j,M}$ the retained set has a number of components bounded independently of M and total length $(\beta - \alpha) + o(M^{-1})$. On each component Φ_M of (5.4) is strictly increasing, and consecutive phase points enclose distinct zeros of $F_M(z(\theta))$; since $\text{Var } \psi$ is bounded (Lemma 3.6),

$$Z_M[\alpha, \beta] \geq \frac{M+1}{\pi}(\beta - \alpha) - O(1).$$

The same bound on the half-open complementary intervals $[h/M, \alpha)$ and $(\beta, \pi/r - h/M]$ gives at least $\frac{M+1}{\pi}(\frac{\pi}{r} - (\beta - \alpha)) - O(1)$ further distinct zeros. As z is monotone, these lie outside $z([\alpha, \beta])$, so their number, counted distinctly, together with $Z_M[\alpha, \beta]$, counted with multiplicity, is at most $\deg F_M = \lfloor M/r \rfloor$ (Lemma 2.3), whence

$$Z_M[\alpha, \beta] \leq \deg F_M - \frac{M+1}{\pi} \left(\frac{\pi}{r} - (\beta - \alpha) \right) + O(1) = \frac{M+1}{\pi}(\beta - \alpha) + O(1).$$

Hence $Z_M[\alpha, \beta] = \frac{M+1}{\pi}(\beta - \alpha) + O(1)$, and dividing by $\deg F_M$,

$$\mu[F_M](z([\alpha, \beta])) \longrightarrow \frac{r}{\pi}(\beta - \alpha) = \mu_\infty(z([\alpha, \beta])).$$

By Proposition 5.1, the retained phase count already supplies $\deg F_M - O(1)$ distinct zeros; the deleted endpoint and amplitude windows of (4.3) and the at most C_B exceptional zeros therefore carry only $O(1)$ zeros, counted with multiplicity, a mass $O(1/M)$ vanishing after normalization. Consequently

$$\mu[F_M](\mathbb{C} \setminus z([\alpha, \beta])) \longrightarrow 1 - \frac{r}{\pi}(\beta - \alpha).$$

Given $\varepsilon > 0$, choose α near 0 and β near π/r with $1 - \frac{r}{\pi}(\beta - \alpha) < \varepsilon$. The limit above gives $\mu[F_M](z([\alpha, \beta])) > 1 - \varepsilon$ for all large M ; each $\mu[F_M]$ is finitely supported, so adjoining the supports of the remaining finitely many measures to the compact set $z([\alpha, \beta])$ yields one compact set serving every M , and $\{\mu[F_M]\}$ is tight. Let μ_* be a subsequential weak limit. Each $z([\alpha, \beta])$ is compact, so the portmanteau inequality for closed sets and the count above give

$$(6.1) \quad \mu_*(z([\alpha, \beta])) \geq \frac{r}{\pi}(\beta - \alpha).$$

A finite measure has at most countably many atoms, so the parameters carrying an atom of μ_* form a countable subset of $(0, \pi/r)$ and their complement is dense; in particular $0 < \alpha_0 < \dots < \alpha_k < \pi/r$ may be chosen with no $z(\alpha_i)$ an atom of μ_* , and with α_0 and α_k arbitrarily close to the ends. Fix $0 < \alpha < \beta < \pi/r$ with $z(\alpha)$ and $z(\beta)$ not atoms, and take α and β as two consecutive α_i . The sets $z([\alpha_{i-1}, \alpha_i])$ then meet only in μ_* -null points, so their masses sum to at most $\mu_*(\mathbb{C}) = 1$, while each excess $\mu_*(z([\alpha_{i-1}, \alpha_i])) - \frac{r}{\pi}(\alpha_i - \alpha_{i-1})$ is nonnegative by (6.1); the excesses therefore sum to at most $1 - \frac{r}{\pi}(\alpha_k - \alpha_0)$. Letting $\alpha_0 \downarrow 0$ and $\alpha_k \uparrow \pi/r$ with α and β held fixed sends that bound to 0, so every excess vanishes:

$$\mu_*(z([\alpha, \beta])) = \frac{r}{\pi}(\beta - \alpha) = \mu_\infty(z([\alpha, \beta])).$$

Such intervals are closed under intersection and generate the Borel subsets of $\overline{I_{Q,r}}$, and their total mass is 1, so $\mu_* = \mu_\infty$; in particular μ_* has no atoms after all. Every subsequential limit being μ_∞ , the full sequence converges weakly. $\square$

Corollary 6.2. *Under the hypotheses of Theorem 1.1,*

$$\frac{\#\{x \in I_{Q,r} : P_m(x) = 0\}}{\deg P_m} \longrightarrow 1$$

along the coefficient polynomials of positive degree, where the numerator counts distinct zeros.

Proof. Theorem 1.1 gives a bounded deficit, while $\deg P_m = m/r + O(1)$ by Remark 2.4. $\square$

Proposition 6.3 (Dependence on the numerator is necessary). *For each admissible denominator $Q(t) + zt^r$, the constant C in Theorem 1.1 cannot be bounded uniformly over all proper numerators N ; in particular, no bound can depend on r and $\deg Q$ alone.*

Proof. Fix any admissible denominator and let $R(z) \in \mathbb{R}[z]$. Taking $N(t, z) = R(z)$ gives $P_m(z) = R(z)H_m(z)$. Choosing

$$R(z) = \prod_{j=1}^L (z + j)$$

places L negative zeros in every nonzero P_m , and L is arbitrary. $\square$

Exceptional zeros need not arise by such a factorization: Figure 1(B) shows the bulk and the exceptional zeros for a numerator that is not such a product.

Remark 6.4. The bound is optimal in order—bounded in m by Theorem 1.1, and necessarily dependent on N by Proposition 6.3; its sharp value is left open (Question 7.1).

7. FURTHER QUESTIONS

The bound of Theorem 1.1 is not explicit in Q, r, N , and the numerators for which P_m has only positive real zeros are not characterized. The closing remark records which features of $Q(t) + zt^r$ the Laurent–Gauss reduction uses, and why it does not extend to linear-type denominators.

Question 7.1. Can the constant in Theorem 1.1 be bounded explicitly in terms of

$$\deg Q, \quad r, \quad \deg_t N, \quad \deg_z N,$$

and the multiplicity of the smallest zero of Q , the orders of the endpoint collisions, and separation data for the zeros of Q ? With $E = \deg_z N$ as in (2.1), the Laurent reduction shows that the relevant numerator complexity is controlled by

$$\deg \left[t^{rE} N \left(t, -\frac{Q(t)}{t^r} \right) \right] \leq \deg_t N + E \max\{\deg Q, r\}.$$

For the pencil $1 + s + t + zst$ with numerator $\tilde{N}(s, t, z)$ such an explicit bound is available: the numerator degrees alone give $\deg_s \tilde{N} + \deg_t \tilde{N} + 2 \deg_z \tilde{N}$ [8].

Question 7.2. Characterize the numerators $N(t, z)$ for which every nonzero P_m has only positive real zeros. This is a numerator-sensitive refinement of the operator problem posed by Forgács and Tran in their Problem 15 [4]. Necessary and sufficient conditions of this kind are known for a linear combination of Chebyshev polynomials with coefficients the powers of $az + b$ [13].

Remark 7.3. The Laurent–Gauss reduction uses two features of $Q(t) + zt^r$: linearity in z , and that the z -coefficient t^r is a unit in $\mathbb{R}[t, t^{-1}]$, which makes the quotient a Laurent polynomial and yields a fixed univariate numerator after a finite shift. For a denominator $Q_0(t) + zt^r Q_1(t)$ with nonconstant Q_1 , the z -coefficient $t^r Q_1$ is no longer a unit: eliminating z requires localizing at Q_1 and produces a fixed rational weight with poles at the zeros of Q_1 , so a finite-shift reduction to a polynomial numerator is no longer automatic. Forgács and Tran study eventual hyperbolicity for such linear-type denominators under separation hypotheses [6]; extending the

bounded-defect theorem there requires a new algebraic reduction together with control of the zeros and poles introduced by Q_1 in the weighted principal amplitude.

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